WEEKLY PROGRESS REPORT – WEEK 3

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

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Week 3 Objectives

The objective of Week 3 was to develop the ETL pipeline, integrate cloud storage and database systems, and automate data ingestion workflows.

Activities Completed

1. **ETL Pipeline Development**

Python and Pandas were used to develop an ETL pipeline for extracting, transforming, and loading healthcare data.

2. **AWS S3 Integration**

AWS S3 was configured as the cloud storage layer for storing healthcare datasets and supporting data ingestion workflows.

3. **Database Configuration**

PostgreSQL and AWS RDS PostgreSQL were configured to store processed healthcare data for analytics and reporting.

4. **Data Loading Process**

Healthcare records were successfully loaded into the database after validation and preprocessing.

5. **Apache Airflow Workflow Automation**

Apache Airflow DAGs were developed to automate ETL execution and workflow orchestration.

6. **Pipeline Testing and Validation**

The ETL pipeline was tested to ensure successful extraction, transformation, and loading of healthcare records.

7. Team Coordination and Review

The team reviewed ETL workflows, validated database loading results, and discussed the next phase of data transformation and analytics.

Deliverables Completed

• ETL pipeline implementation

• AWS S3 integration

• PostgreSQL database setup

• AWS RDS PostgreSQL integration

• Apache Airflow DAG creation

• Automated data loading workflow

• ETL validation report

Challenges Faced

• Configuring cloud storage integration

• Database connectivity setup

• Workflow automation configuration

• Validating ETL execution results

Plan for Week 4

• Implement PySpark processing

• Design Data Warehouse

• Build Star Schema

• Develop SQL analytics layer

• Optimize analytical queries

Conclusion

Week 3 focused on ETL pipeline development, cloud integration, database loading, and workflow automation. The healthcare data pipeline was successfully automated using Apache Airflow and integrated with AWS cloud services.